

Supplementary Information for “RDMA: Cost Effective Agent-Driven Rare Disease Mining from Electronic Health Records”

Inference and Hardware Cost Estimates

Here, we describe our hardware and inference costs within the context of our benchmark statistics. While local hardware costs are based directly on the listing available for commonplace workstations, we follow a simple equation to compute our cost metrics in benchmarking phenotype extraction in Supplementary Tables 1 and 4. We discuss local hardware costs to run these systems in Supplementary Table 2. For cost reasons, our downstream performance benchmarking was done with the 4-bit quantized variants of each model when the model sizes are greater than 8B parameters. We only benchmark costs for non-quantized variants for the Llama 3.3 70B as our largest locally deployable model. Quantization was done with the bitsandbytes package [1]. We benchmark primarily on our case study cohort (CSC) benchmark.

GPU	Rental Cost (\$/hr)
A6000	0.5
RTX 3090	0.1

Supplementary Table 1: GPU rental costs per hour.

Category	GPU Configuration	Estimated Cost
Very Low	N/A	\$120
Low	1×3090	\$2,200
Medium	1×A6000	\$6,520
High	4×A6000	\$38,500

Supplementary Table 2: **Local Hardware Cost Categories.** These cost categories are referenced in the performance comparison in our discussion. We note that the approximate costs are based from current workstation prices [2] [3] [4] as of April 26, 2025. In principle, rule-based approaches do not need a GPU to run scalably where even a Raspberry Pi is computationally sufficient.

Variable Name	Statistic	Total
TN	Total Notes	331,794
MNL	Median MIMIC-IV Note Length (word count)	1,320
MCSL	Benchmark Median Case Study Length (word count)	271.5

Supplementary Table 3: MIMIC4 Notes Statistics.

Baseline	GPU	Run Time (m)	Cost Calculation
RAG-HPO (Mistral 24B ^q)	1×RTX 3090	39	$\frac{39 \times 0.1}{60} \times \frac{\text{MNL} \times \text{TN}}{\text{MCSL}}$
RAG-HPO (Llama 3.3-70B ^q)	1×A6000	62	$\frac{62 \times 0.5}{60} \times \frac{\text{MNL} \times \text{TN}}{\text{MCSL}}$
RAG-HPO (Llama 3.3-70B)	4×A6000	70	$\frac{70 \times 4 \times 0.5}{60} \times \frac{\text{MNL} \times \text{TN}}{\text{MCSL}}$
RDMA	1×RTX 3090	121	$\frac{121 \times 0.1}{60} \times \frac{\text{MNL} \times \text{TN}}{\text{MCSL}}$

Supplementary Table 4: Baseline comparison of different methods with their runtime and cost calculations.

Supplementary Table 5: Runtime comparison on CSC dataset using Mistral 24B 4-bit quantized. Given that the CSC dataset contains 32,260 words, we have a pipeline that can extract, verify, and match HPO entities at a rate of approximately 278 words a minute on an RTX 3090.

Method	Extraction (m)	Verification (m)	Matching (m)
RAG HPO	15	—	24
RDMA	68	29	24

Supplementary Table 6: Statistics of rare disease NER benchmarks and phenotype datasets used for evaluation. RareDis splits: train (711 docs, 3,608 entities, 111,369 words), dev (97 docs, 525 entities, 15,268 words), test (203 docs, 1,088 entities, 31,042 words). For BiolarkGSC, our phenotype training corpus, we use a random 80-10-10 split, roughly resulting in 182 train / 23 dev / 23 test documents. For all general LLM-based baselines, we use the entire corpus for benchmarking as they do not require training.

Category	Dataset	Docs	Entities	Total Words	Avg. Words	Max Words
Phenotype	BioLarkGSC [5]	228	2,773	33,942	148.9	359
	CSC [6]	116	1,813	32,260	278.1	675
Rare Disease	RareDis [7]	1,011	5,221	157,679	156.0	601
	MIMIC3-RD [8]	117	176	221,980	1,897.3	6,726

Performance Metric Calculations

For each clinical document x_i , we compare the set of ground-truth human-annotated codes with the set of predicted codes from our RDMA framework. Let $\Phi_P(x_i)$ and $\Phi_R(x_i)$ denote the ground-truth phenotype and rare disease codes for document x_i , respectively, and let $\hat{\Phi}_P(x_i)$ and $\hat{\Phi}_R(x_i)$ denote the corresponding predicted codes.

For each document x_i and each task (phenotypes or rare diseases), we define:

True Positives:

$$TP_P^i = |\hat{\Phi}_P(x_i) \cap \Phi_P(x_i)| \quad (1)$$

$$TP_R^i = |\hat{\Phi}_R(x_i) \cap \Phi_R(x_i)| \quad (2)$$

False Positives:

$$FP_P^i = |\hat{\Phi}_P(x_i) \setminus \Phi_P(x_i)| \quad (3)$$

$$FP_R^i = |\hat{\Phi}_R(x_i) \setminus \Phi_R(x_i)| \quad (4)$$

False Negatives:

$$FN_P^i = |\Phi_P(x_i) \setminus \hat{\Phi}_P(x_i)| \quad (5)$$

$$FN_R^i = |\Phi_R(x_i) \setminus \hat{\Phi}_R(x_i)| \quad (6)$$

We then aggregate these counts across all n documents in our corpus to compute overall performance metrics:

$$TP_P = \sum_{i=1}^n TP_P^i, \quad FP_P = \sum_{i=1}^n FP_P^i, \quad FN_P = \sum_{i=1}^n FN_P^i \quad (7)$$

$$TP_R = \sum_{i=1}^n TP_R^i, \quad FP_R = \sum_{i=1}^n FP_R^i, \quad FN_R = \sum_{i=1}^n FN_R^i \quad (8)$$

For phenotype extraction:

$$\text{Precision}_P = \frac{TP_P}{TP_P + FP_P} \quad (9)$$

$$\text{Recall}_P = \frac{TP_P}{TP_P + FN_P} \quad (10)$$

$$\text{F1}_P = 2 \cdot \frac{\text{Precision}_P \cdot \text{Recall}_P}{\text{Precision}_P + \text{Recall}_P} \quad (11)$$

For rare disease extraction:

$$\text{Precision}_R = \frac{TP_R}{TP_R + FP_R} \quad (12)$$

$$\text{Recall}_R = \frac{TP_R}{TP_R + FN_R} \quad (13)$$

$$\text{F1}_R = 2 \cdot \frac{\text{Precision}_R \cdot \text{Recall}_R}{\text{Precision}_R + \text{Recall}_R} \quad (14)$$

Supplementary Table 7: Phenotype extraction results. $F1^\pm$ denotes the half-width of the 95% bootstrap CI. BioLARK-GSC: $n=228$ docs; CSC: $n=116$ docs. Zeroshot LLMs score 0 on both datasets (cannot generate HPO identifiers without structured prompting). Best per approach group **bolded**.

Approach	Model	BioLarkGSC+	CSC
<i>Baselines</i>			
	BioBERT-MRC	0.498 ± 0.064	0.382 ± 0.030
	BioClinicalBERT	0.441 ± 0.081	0.392 ± 0.031
	FastHPOCR	0.603 ± 0.023	0.480 ± 0.029
	Dictionary	0.326 ± 0.025	0.319 ± 0.025
	i2b2	0.491 ± 0.024	0.461 ± 0.031
	PhenoGPT	0.427 ± 0.027	0.373 ± 0.035
<i>Zeroshot</i>			
	GPT-5	$0.000 \pm .000$	$0.000 \pm .000$
	Llama-3 70B	$0.000 \pm .000$	$0.000 \pm .000$
	Llama-3 8B	$0.000 \pm .000$	$0.000 \pm .000$
	Mistral 24B	$0.000 \pm .000$	$0.000 \pm .000$
	Qwen3 32B	$0.000 \pm .000$	$0.000 \pm .000$
<i>RAG</i>			
	GPT-5	0.525 ± 0.027	0.543 ± 0.027
	Llama-3 70B	0.474 ± 0.026	0.624 ± 0.025
	Llama-3 8B	0.345 ± 0.024	0.509 ± 0.033
	Mistral 24B	0.398 ± 0.026	0.595 ± 0.034
	Qwen3 32B	0.401 ± 0.024	0.594 ± 0.030
<i>RDMA</i>			
	GPT-5	0.545 ± 0.029	0.646 ± 0.041
	Llama-3 70B	0.484 ± 0.023	0.627 ± 0.027
	Llama-3 8B	0.331 ± 0.019	0.519 ± 0.033
	Mistral 24B	0.559 ± 0.021	0.657 ± 0.029
	Qwen3 32B	0.473 ± 0.022	0.649 ± 0.028

Bootstrapped Confidence Intervals

Because reruns are prohibitively expensive, we minimize LLM temperature (i.e $t=0.001$) to limit variance. However, we still include bootstrapped confidence intervals (Supplementary Tables 7 and 8).

To estimate the stability of micro-F1 scores without rerunning experiments, we apply document-level bootstrap resampling over per-document evaluation counts (tp_i, fp_i, fn_i) drawn from each evaluated run.

Point Estimate. Micro-F1 is computed from aggregate counts over all N documents:

$$\begin{aligned}
 TP &= \sum_{i=1}^N tp_i, & FP &= \sum_{i=1}^N fp_i, & FN &= \sum_{i=1}^N fn_i, \\
 P &= \frac{TP}{TP + FP}, & R &= \frac{TP}{TP + FN}, & F_1 &= \frac{2PR}{P + R},
 \end{aligned} \tag{15}$$

with each term defined as zero when its denominator is zero.

Supplementary Table 8: Rare disease mention extraction results. $F1^\pm$ denotes the half-width of the 95% bootstrap CI. MIMIC3-RD Entity: $n=117$ docs; MIMIC3-RD Code: $n=79$ docs; RareDis: $n=1,011$ docs. Best per approach group **bolded**.

Approach	Model	MIMIC-III (text)	MIMIC-III (code)	RareDis
<i>Baselines</i>				
	BioBERT-MRC	0.046 ± 0.011	0.046 ± 0.012	0.717 ± 0.027
	BioClinicalBERT	0.050 ± 0.011	0.050 ± 0.013	0.673 ± 0.032
	Dictionary	0.431 ± 0.074	0.384 ± 0.066	0.555 ± 0.022
	i2b2	0.020 ± 0.004	0.019 ± 0.004	0.260 ± 0.012
<i>Zeroshot</i>				
	GPT-5	0.497 ± 0.069	0.098 ± 0.056	0.810 ± 0.016
	Llama-3 70B	0.223 ± 0.038	0.000 ± 0.000	0.705 ± 0.019
	Llama-3 8B	0.134 ± 0.024	0.000 ± 0.000	0.548 ± 0.020
	Mistral 24B	0.228 ± 0.041	0.002 ± 0.004	0.704 ± 0.020
	Qwen3 32B	0.326 ± 0.066	0.000 ± 0.000	0.651 ± 0.022
<i>RAG</i>				
	GPT-5	0.505 ± 0.069	0.391 ± 0.068	0.826 ± 0.017
	Llama-3 70B	0.125 ± 0.023	0.139 ± 0.024	0.483 ± 0.018
	Llama-3 8B	0.047 ± 0.012	0.025 ± 0.007	0.267 ± 0.015
	Mistral 24B	0.592 ± 0.068	0.032 ± 0.010	0.226 ± 0.012
	Qwen3 32B	0.133 ± 0.027	0.143 ± 0.030	0.748 ± 0.019
<i>RDMA</i>				
	GPT-5	0.513 ± 0.072	0.465 ± 0.083	0.780 ± 0.016
	Llama-3 70B	0.442 ± 0.056	0.439 ± 0.069	0.845 ± 0.013
	Llama-3 8B	0.072 ± 0.013	0.097 ± 0.020	0.595 ± 0.016
	Mistral 24B	0.592 ± 0.068	0.526 ± 0.080	0.814 ± 0.014
	Qwen3 32B	0.528 ± 0.057	0.467 ± 0.071	0.828 ± 0.013

Resampling Procedure. We draw $B = 1,000$ bootstrap replicates. In each replicate, N documents are sampled with replacement from the original N ; micro-F1 is recomputed from the resampled aggregate counts, yielding a distribution $\{F_1^{(b)}\}_{b=1}^B$.

Confidence Interval. A $(1-\alpha)$ confidence interval is read from the sorted bootstrap distribution. For a 95% CI ($\alpha = 0.05$):

$$\hat{F}_1^{\text{lo}} = F_1^{(\lfloor \alpha/2 \cdot B \rfloor)}, \quad \hat{F}_1^{\text{hi}} = F_1^{(\lfloor (1-\alpha/2) \cdot B \rfloor)}, \quad (16)$$

corresponding approximately to the 2.5th and 97.5th percentiles. All runs with fewer than 20 documents are excluded to avoid partial or debug evaluations.

Rationale. Resampling is performed at the document level rather than the entity or token level, as the clinical note or disease case is the natural unit of variance: some documents are substantially harder than others, and token-level resampling would underestimate uncertainty by treating all mentions as independent.

AI Assisted Corrections and Annotations on MIMIC-III Corpus

We showcase a variety of cases from poor annotations from the original dataset in Supplementary Fig. 1 and our own error analysis of the 4th dataset refinement step in RDMA in Supplementary Fig. 2.

Entity Reviewfalse negatives

Basic Information
Entity:
NPH
Document ID:
47713
ORPHA Code:
314928
Current Status:
Not marked as rare disease

Top Candidate
Name:
non-acquired pituitary hormone deficiency
ID:
ORPHA:95488
Similarity:
66.9%

Text Context
crackers. Her insulin regimen was made less aggressive on admission to the floor with an regimen of **NPH** [**12-2**] in the am/pm and sliding scale insulin as well. She felt that she was not eating as much

Flag Reason
low_orpha_similarity [FLAGGED: Entity verified as not a rare disease despite being categorized as false negative]

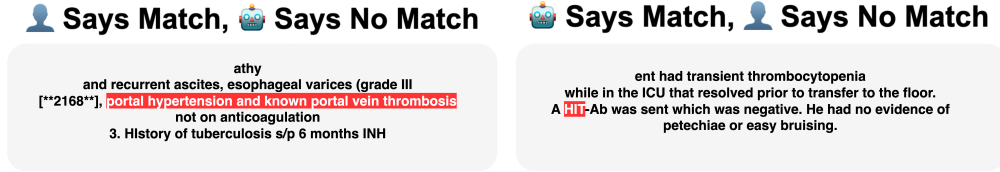
Your Decision

You marked this as NOT a rare disease

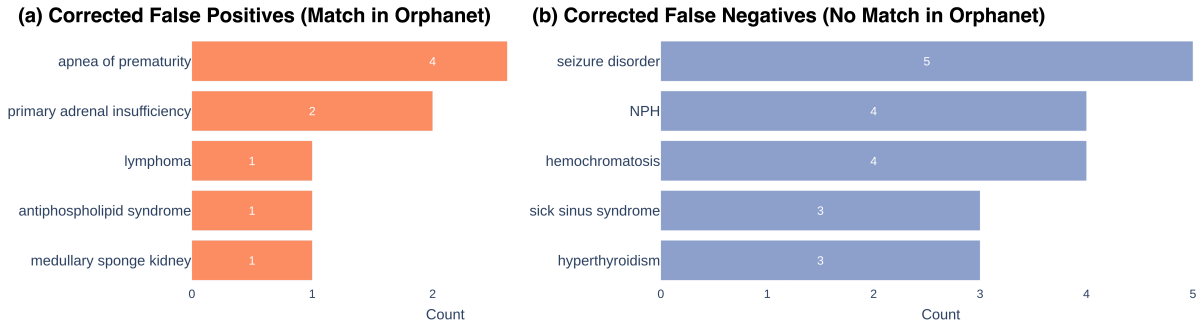
Change Decision

Supplementary Fig. 1: **Example of Inappropriate Annotation in Noisy Dataset.** We note that while NPH as an abbreviation can be related to “normal pressure hydrocephalus” or other related conditions in the Orphanet ontology as annotated by [9], NPH here in this context is actually referring to neutral protamine hagedorn, a type of insulin used to treat diabetes.

Examples of RDMA Flagged Entities in Dataset Refinement. We investigate which entities were primarily flagged by RDMA for human review. We visualize the top cases in Supplementary Fig. 3, where we observe that many entities mined by [9] were indeed not rare diseases, while several key rare disease entities were missed in their original annotations.



Supplementary Fig. 2: **Failure Modes of RDMA in Rare Disease Mining.** When reviewing existing annotations, we showcase two cases where our human annotator disagrees with RDMA’s refinement step. Specifically, we see that our refinement agent fails to classify portal vein thrombosis as a rare disease, primarily because its Orphanet listing “Non-cirrhotic and non-tumoral portal vein thrombosis” does not exactly match the entity “portal vein thrombosis”. On the other hand, while the agent is able to classify “HIT” as “heparin induced thrombocytopenia”, it does not capture the context “which was negative” properly.



Supplementary Fig. 3: **Top 5 Corrected Annotations with RDMA.** RDMA effectively assists human annotation by identifying errors in the noisy historical dataset from [9]. The figure highlights the top 5 corrections, including both false negatives (a) such as seizure disorder incorrectly labeled as a rare disease in Orphanet, and false positives (b) that were missed in the initial annotations. Of the 72 annotations flagged by RDMA for human review, 55 (76.4%) were correctly identified as erroneous, demonstrating its value as an annotation assistant for prioritizing human supervision. While hemochromatosis is typically considered a rare condition, in ICU contexts such as MIMIC notes, our physician deemed that to not necessarily be the case. Furthermore, Orphanet [10] makes a distinction between **rare** hemochromatosis and hemochromatosis.

Annotator Guidelines

We build on the rare disease mention annotations from [9] in MIMIC-III, first filtering out clearly erroneous annotations as detailed in Supplementary Table 10. Following filtering, a Ph.D. student reviewed each remaining entity from [9] under physician guidance, applying the review process outlined in Supplementary Fig. 4 to determine whether each mention directly or indirectly implies a rare disease. This produced our final curated annotation set.

For the RDMA-assisted annotation run, the same Ph.D. student conducted a separate review pass over only the entities flagged by RDMA for human review, applying a binary correction, marking each as correct or incorrect. RDMA flags entities exhibiting disagreement across two independent mining passes and a final agent verification check, meaning the annotator is presented only with the subset of entities where automated agreement could not be established.

Inter-Annotator Agreement. We measure agreement between the human annotator and RDMA using Cohen’s κ , which corrects observed agreement by chance agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (17)$$

Given a 2×2 confusion matrix over $N = \text{TP} + \text{TN} + \text{FP} + \text{FN}$ binary judgments, observed agreement and chance agreement are computed as:

$$p_o = \frac{\text{TP} + \text{TN}}{N} \quad (18)$$

$$p_e = p_{\text{human}} \cdot p_{\text{RDMA}} + (1 - p_{\text{human}}) \cdot (1 - p_{\text{RDMA}}) \quad (19)$$

where $p_{\text{human}} = (\text{TP} + \text{FN})/N$ and $p_{\text{RDMA}} = (\text{TP} + \text{FP})/N$ are the marginal rates of each rater labeling an entity as a rare disease mention. We report κ under the Landis & Koch interpretation bands, where $\kappa \in [0.41, 0.60]$ indicates moderate agreement and $\kappa \in [0.81, 1.00]$ indicates almost perfect agreement. Agreement between the human annotator and the unassisted prior annotations yielded $\kappa = 0.454$ (moderate), while agreement between the human annotator and RDMA-assisted annotations yielded $\kappa = 0.810$ (almost perfect), demonstrating a substantial improvement in annotation quality under the RDMA-assisted workflow.

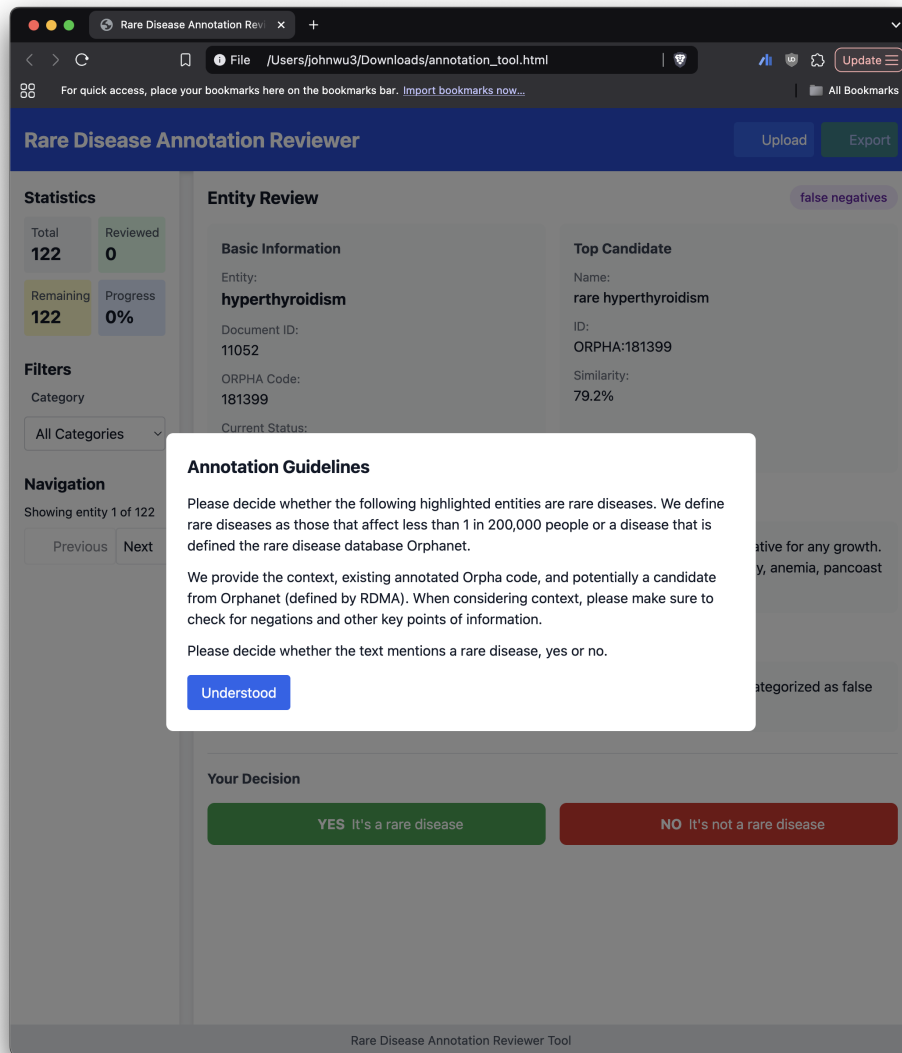
Sentence Agglomeration for Faster Entity Extraction

To optimize the performance of our retrieval-enhanced entity extraction process, we implemented a sentence agglomeration strategy that combines shorter sentences to reduce computational costs.

Sentence agglomeration with a minimum size of 500 characters reduced processing time by 75% with a modest trade-off in extraction quality, making this approach particularly valuable for time-sensitive applications or large document collections.

Algorithm 1 Sentence Agglomeration Algorithm

```
1: procedure MERGESMALLSENTENCES(sentences, min_size)
2:   if sentences is empty then
3:     return empty list
4:   end if
5:   if min_size is null or min_size  $\leq 0$  then
6:     return sentences unmodified
7:   end if
8:   merged_sentences  $\leftarrow$  empty list
9:   current_idx  $\leftarrow 0$ 
10:  while current_idx  $< |sentences|$  do
11:    current_sentence  $\leftarrow sentences[current\_idx]$ 
12:    if  $|current\_sentence| \geq min\_size$  then
13:      Append current_sentence to merged_sentences
14:      current_idx  $\leftarrow current\_idx + 1$ 
15:      continue
16:    end if
17:    merged_chunk  $\leftarrow current\_sentence$ 
18:    next_idx  $\leftarrow current\_idx + 1$ 
19:    while next_idx  $< |sentences|$  and  $|merged\_chunk| < min\_size$  do
20:      if merged_chunk is not empty and sentences[next_idx] is not empty then
21:        merged_chunk  $\leftarrow merged\_chunk + " " + sentences[next\_idx]$ 
22:      else
23:        merged_chunk  $\leftarrow merged\_chunk + sentences[next\_idx]$ 
24:      end if
25:      next_idx  $\leftarrow next\_idx + 1$ 
26:    end while
27:    Append merged_chunk to merged_sentences
28:    current_idx  $\leftarrow next\_idx$ 
29:  end while
30:  return merged_sentences
31: end procedure
```



Supplementary Fig. 4: **Annotator Guidelines.** Annotators are asked if the mention in text directly or indirectly implies a rare disease.

Metric	No Agglomeration	With Agglomeration (Min Size = 500)
Total documents analyzed	117	117
Average word count per document	1897	1897
Precision	0.89	0.82
Recall	0.44	0.42
F1 score	0.59	0.55
Extraction run time (hours)	8:36	2:09

Supplementary Table 9: **Comparative Performance Metrics for Different Sized Text Chunks in RDMA for the Entity Extraction Step.** We observe that performance declines slightly, but we are able to extract entities at a substantially higher rate (4x faster) from over 200,000 words.

MIMIC3-RD Preliminary Filtering

Preliminary Filtering. Before applying our RDMA refinement (Supplementary Fig. 1), we filter out known incorrect keywords and add known rare disease mentions, as detailed in Supplementary Table 10. These changes significantly impact the dataset statistics (Supplementary Table 11), reducing annotations from 1,073 to 333. We use this filtered dataset for our case study in the main manuscript.

Kept Abbreviations	HIT, ALS, and NPH
Manually Removed Terms	Hyperlipidemia, dyslipidemia, hypercholesterolemia
Manually Added Terms	papillary carcinoma, glioblastoma multiforme, transitional cell carcinoma, multifocal atrial tachycardia (mat), sarcoidosis, methemoglobinemia, central nervous system and systemic lymphoma, sclerosis cholangitis, mediastinitis, mesenteric vein thrombosis, multiple myeloma, hepatocellular carcinoma, primary cns lymphoma, sclerosing cholangitis, bechet’s disease, neovascular glaucoma, meningocele, alopecia, neovascular glaucoma angle closure, pyoderma gangrenosum, budd-chiari, intraductal papillary mucinous tumor, complex tracheal stenosis, cervical stenosis, bronchiectasis, medullary sponge kidney, protein s, antiphospholipid antibody syndrome, protein c, hepatocellular ca, acute myelogenous leukemia, anaplastic thyroid carcinoma, thymoma, congenital bleeding disorder, tracheal stenosis

Supplementary Table 10: **Initial Keyword-based Filtering Attempts** Our clinician identified numerous incorrectly annotated terms from the original entity set. For instance, "MR" was incorrectly tagged as "Multicentric reticulohistiocytosis" in [9]’s annotations, though it commonly refers to "magnetic resonance". Our clinician flagged all abbreviations except HIT, ALS, and NPH. We also removed common disease terms like hyperlipidemia, as their rare variants are specifically defined differently in the Orphanet ontology. Finally, our clinician manually added the terms listed above. For each added term, we checked its presence in each document and, if found, added an annotation with its corresponding ORPHA code.

	Original	After Initial Filtering
Number of Notes	312	117
Number of Annotations	1,073	333

Supplementary Table 11: **Rare Disease Annotation Statistics Before and After Initial Keyword Filtering.** We eliminated over 700 misannotated terms before conducting evaluations in the main manuscript.

Prompts

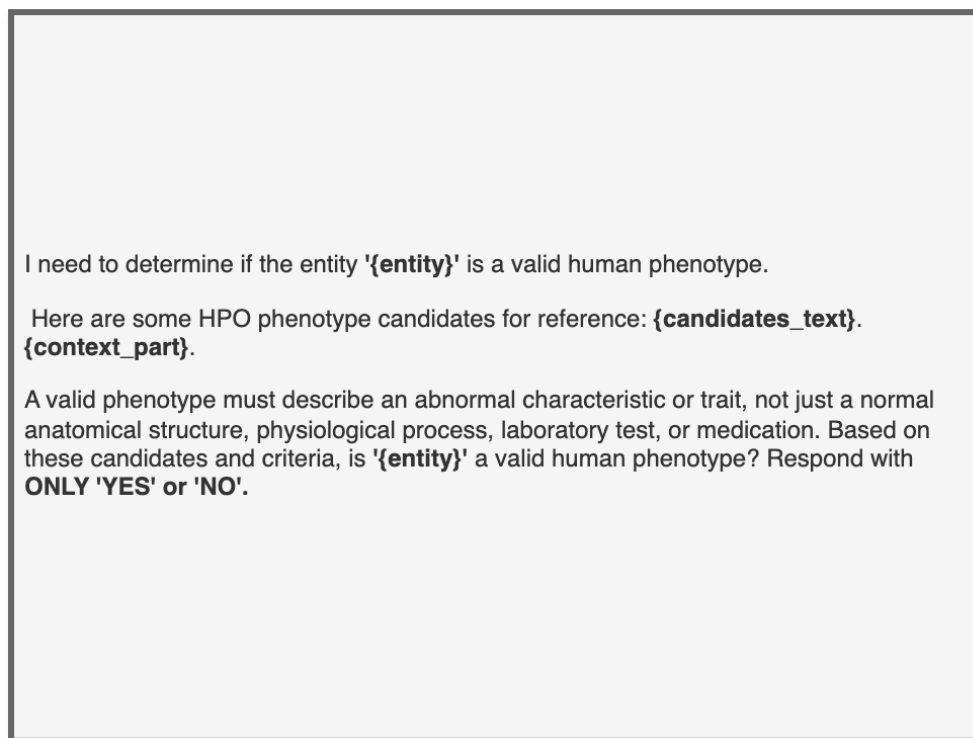
We showcase all of the LLM prompts used in RDMA below.

I have a clinical sentence: "{sentence}" Here are some relevant HPO terms for context that are potentially within the sentence: {context_text} Based on this sentence and the provided HPO terms for context, extract all phenotype terms (genetic inheritance patterns, anatomical anomalies, clinical symptoms, diagnostic findings, test results, conditions or syndromes) from the sentence. IGNORE NEGATIVE FINDINGS, NORMAL FINDINGS, AND ANY TERMS MENTIONED IN FAMILY HISTORY. Please include the full term and any additional context that is part of the term. MAKE SURE IT MATCHES EXACTLY AS IT APPEARS IN THE SENTENCE. Return the extracted terms as a JSON object with a single key 'findings', which contains the list of extracted terms spelled correctly.	I have CLINICAL TEXT: "{sentence}" Here are some relevant ORPHA rare disease terms for reference that may help you find rare disease mentions in the sentence: {context_text} Based on this sentence and the provided rare disease terms as reference, extract all potential disease mentions that are NOT negated (i.e., NOT preceded by 'no', 'not', 'without', 'ruled out', etc.). Please also include any potential abbreviations that might be referring to rare diseases in the CLINICAL TEXT. Return only a Python list of strings, with each disease exactly as it appears in the CLINICAL TEXT. Ensure the output is concise without any additional notes, commentary, or meta explanations.
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Supplementary Fig. 5: **Entity Extraction Prompts.** We showcase both HPO extraction (left) and Rare Disease extraction (right) prompts here.

HPO Verification and Matching Prompts

We showcase all of the prompts used for HPO extraction here.



Supplementary Fig. 6: **Verifying an Entity is a Phenotype.** This reasoning step is used repeatedly in verifying all phenotype implications, whether done by a lab test or a generated implication.

I need to determine if the entity '{entity}' contains information about a laboratory test with a measured value. {context_part}. Laboratory tests with measured values include examples like:

- 'Hemoglobin 8.5 g/dL'
- 'Elevated white blood cell count of 15,000/ μ L'
- 'Sodium 140 mEq/L'

Does '{entity}' represent a lab test with a numerical value/result? Respond with ONLY 'YES' or 'NO'.

Supplementary Fig. 7: **Lab Test Check**. LLMs are asked to see if an entity is indicative of a lab test.

Analyze this potential laboratory test entity: '{entity}'. {context_part} {reference_part}
{sample_part}

Extract the lab test name, value (with units if available), and determine if the result is abnormal. If abnormal, provide a clear medical description of the abnormality (e.g., 'elevated glucose', 'leukopenia').

Provide your response in this EXACT JSON format: {
"lab_name": "[extracted lab test name]",
"value": "[extracted value with units if available]",
"units": "[extracted units if separable from value]",
"is_abnormal": true/false,
"abnormality": "[descriptive term for the abnormality, or 'normal' if not
abnormal]",
"direction": "[high/low/normal]",
"confidence": [0.0-1.0 value indicating your confidence]
} Return ONLY the JSON with no additional text.

Supplementary Fig. 8: **Lab Test Implication.** LLMs are asked to determine if a lab event indicates an abnormality in the patient.

I need to determine if the phenotype '{phenotype}' is a valid medical concept.

Here are some HPO phenotype candidates for reference: {candidates_text}. Is '{phenotype}' a valid phenotype in clinical medicine? Consider both potential matches in the candidates and your general knowledge of medical phenotypes. Respond with ONLY 'YES' or 'NO'.

Supplementary Fig. 9: **Variant of Phenotype Verification.** This prompt is used to double check if an implied lab test phenotype is within the HPO ontology.

I need to determine if '{entity}' DIRECTLY AND UNAMBIGUOUSLY implies a specific phenotype. Be extremely conservative - only say YES if the implication is clear and specific. {context_part} {retrieval_part} Laboratory values, medications, or procedures DO NOT imply phenotypes unless there is explicit abnormality mentioned. If you're uncertain or the implication requires multiple assumptions, say NO.

Does '{entity}' directly imply a specific phenotype? Respond with ONLY 'YES' if it directly implies a phenotype or 'NO' if it doesn't.

Supplementary Fig. 10: **Entity Implies Phenotype Check.** LLM is asked to check if an entity implies the possibility of a phenotype.

The term '{entity}' might imply a phenotype. {context_part} {retrieval_part} What specific phenotype is directly implied by '{entity}'?

For example, 'hemoglobin of 8 g/dL' implies 'anemia'. I

f you cannot identify a specific phenotype that is DIRECTLY implied with high confidence, respond with EXACTLY 'NO_CLEAR_PHENOTYPE_IMPLIED'.

Provide ONLY the name of the implied phenotype, without any explanation, or 'NO_CLEAR_PHENOTYPE_IMPLIED' if none is clear.

Supplementary Fig. 11: **Phenotype Implication Generation.** The LLM is asked to generate or predict what the implied phenotype is based on context and the entity.

I need to validate whether the following implication is reasonable:

Original entity: '{entity}'
Implied phenotype: '{implied_phenotype}'

{context_part} Be extremely critical and conservative. Say YES only if there is an unambiguous, direct clinical connection between the entity and the proposed phenotype. The connection must be evident from the entity itself, not inferred from general knowledge.

Is this a valid and reasonable implication? Respond with ONLY 'YES' or 'NO'.

Supplementary Fig. 12: **Phenotype Implication Reasoning Check.** The LLM is then asked to double check its implication to make sure it makes sense.

Identify the most appropriate Human Phenotype Ontology (HPO) term for the given patient data and additional context provided. Prioritize terms that are concise and directly relevant to the primary symptom or condition described. Focus on the core subject of each phrase and avoid selecting options with additional descriptive or situational details unless they are essential for accurately capturing the phenotype. Ensure the chosen HPO term closely matches the patient's condition as described, without adding any new or extraneous terms. If multiple phenotypes are present, select and return the single most pertinent HPO term that best represents the primary condition or symptom. Provide only the HPO term itself, with no extra context or commentary.

Query: {entity}

Original Sentence: {original_sentence}

Context: The following related information is available to assist in determining the appropriate HPO terms:

{context_items (joined by newlines)}

Supplementary Fig. 13: **Phenotype Matching**. Finally, the LLM is asked to match the an entity or implied entity to a matching phenotype.

Rare Disease Verification and Matching Prompts

You are a medical expert specializing in rare diseases with comprehensive knowledge of the ORPHANET database. Your task is to determine if a given medical term is semantically equivalent to any of the ORPHA entries provided. For a match to be valid, the entities must refer to the same specific rare disease or syndrome, not just similar conditions.

I need to determine if the term '{entity}' is among any of these rare diseases from ORPHANET:

{entities_text}

Context around entity:
{context}

Decide if '{entity}' is the same disease as any of these entries. Consider synonyms, abbreviations, and variant names. Account for spelling variations and different naming conventions for the same disease entity.

For variants of common diseases, it must be explicitly marked as a rare variant. If there is a partial match, i.e. cholangitis vs. sclerosing cholangitis. There must be a mention of its descriptor (sclerosing) in the term or context itself, otherwise it's an invalid match.

Respond with ONLY '**YES**' if there is a match, and '**NO**' if there is no match.

Supplementary Fig. 14: **Rare Disease Entity Ontology Check.** This prompt checks if the entity is within the Orphanet ontology.

Performance Across Annotation Sets

We disclose what performance metrics look like across the original noisy set of annotations and our different variants of corrected annotations here.

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You are a medical expert specializing in rare diseases with comprehensive knowledge of the ORPHANET database. Your task is to determine if a given medical term is semantically equivalent to any of the ORPHA entries provided. For a match to be valid, the entities must refer to the same specific rare disease or syndrome, not just similar conditions.

I need to determine if the term '{entity}' is among any of these rare diseases from ORPHANET:

{entities_text}

Context around entity:

{context}

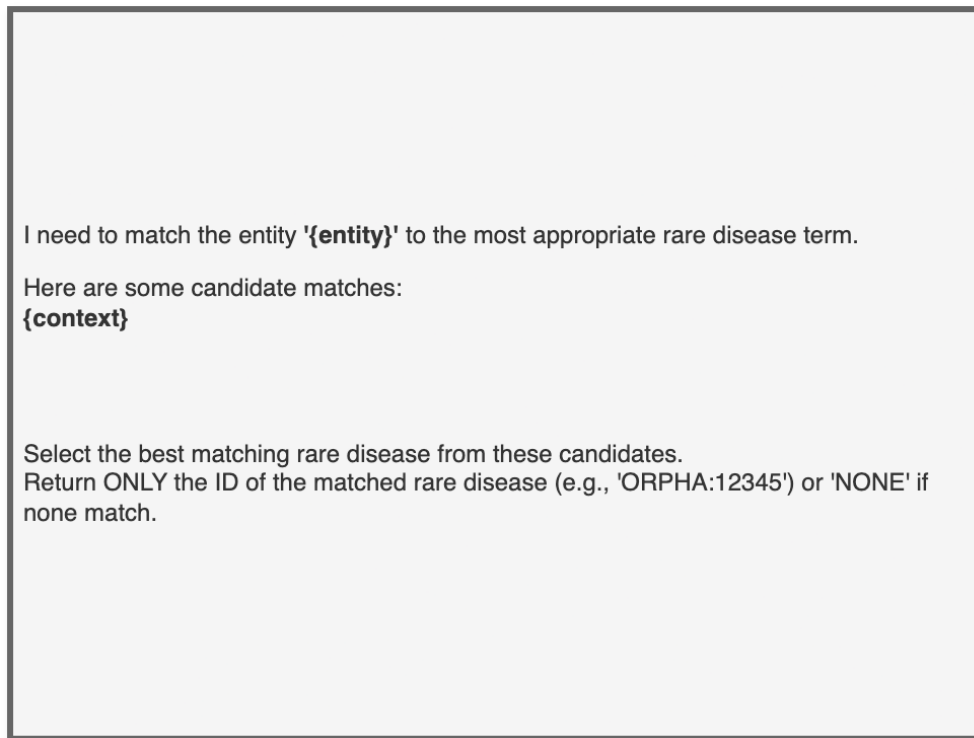
Decide if '{entity}' is the same disease as any of these entries. Consider synonyms, abbreviations, and variant names. Account for spelling variations and different naming conventions for the same disease entity.

For variants of common diseases, it must be explicitly marked as a rare variant. If there is a partial match, i.e. cholangitis vs. sclerosing cholangitis. There must be a mention of its descriptor (sclerosing) in the term or context itself, otherwise it's an invalid match.

Respond with ONLY '**YES**' if there is a match, and '**NO**' if there is no match.

Supplementary Fig. 15: **Rare Disease Check**. This prompt checks if the entity is actually disease, because not all Orphanet entities are diseases.

- [5] Yang, J., Liu, C., Deng, W., Wu, D., Weng, C., Zhou, Y., Wang, K.: Enhancing Phenotype Recognition in Clinical Notes Using Large Language Models: PhenoBCBERT and PhenoGPT (2023). <https://arxiv.org/abs/2308.06294>
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Supplementary Fig. 16: **Rare Disease Matching.** This prompt matches the verified entity to an Orpha code.

Baseline	Model	Precision	Recall	F1
0-shot	Llama 3.3 70B ^q	0.01	0.03	0.02
Retrieve and String Match	MedEmbed	0.23	0.36	0.28
RAG-RD	Mistral 24B ^q	0.17	0.45	0.24
RDMA	Mistral 24B ^q	0.67	0.28	0.39
Human Corrected Labels				
0-shot	Llama 3.3 70B ^q	0.01	0.05	0.02
Retrieve and String Match	MedEmbed	0.25	0.43	0.32
RAG-RD	Mistral 24B ^q	0.14	0.54	0.22
RDMA	Mistral 24B ^q	0.66	0.38	0.48

Supplementary Table 12: **Rare Disease Extraction Performance Comparison.** We showcase performance on the other 2 sets of rare disease mention labels, the noisy original and the human only corrected labels. We observe that RDMA has significantly higher precision than its baselines, and that it overall outperforms all of its baselines in F1. **Bold** denotes best performance.